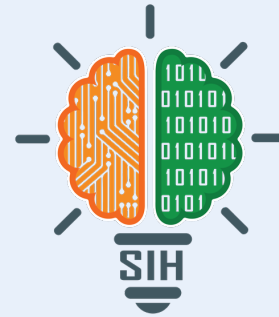


TITLE PAGE

SMART INDIA HACKATHON 2026

Problem Statement ID	SIH26158
Problem Statement Title	Single-Pass Drone Video to Accurate 3D Model Generation System
Theme	Robotics and Drones
PS Category	Software
Organisation	National Technical Research Organisation (NTRO)
Team ID	
Team Name	sixth harmony
Idea Title	OnePass: measurement-grade 3D from a single drone flight



**SMART INDIA
HACKATHON
2026**

PROPOSED SOLUTION (DESCRIBE YOUR IDEA/SOLUTION/PROTOTYPE)

OnePass

Measurement-grade 3D from a single drone flight

In plain terms: fly a drone over an area **once**, and OnePass turns that one video into an accurate, map-ready 3D model. No return trips, no markers placed on the ground first.

WHAT IT DOES

- Builds a 3D model of terrain, buildings, roads and vegetation from one ordinary flight video
- Produces three tiers from that one flight: a fast rough preview, a realistic 3D scene, and a survey-accurate model
- Places the model at its real-world coordinates using the drone's own GPS and altitude sensor

WHY THIS MATTERS

- Standard drone mapping needs several overlapping flights and physical ground markers placed beforehand (ref. 10)
- In a disaster zone or restricted area, teams usually get one flight and no time to place markers

BUILT SO FAR **TIER 1 RUNNING** — photos or video + telemetry → GPS-aligned coloured point cloud in ~30 s, fully offline, with per-point confidence and a browser viewer. Tiers 2–3 and measurement tools are **not** built. Details on slide 3.

A single pass cannot observe every surface. We separate measured geometry from inferred geometry and let the operator see which is which.

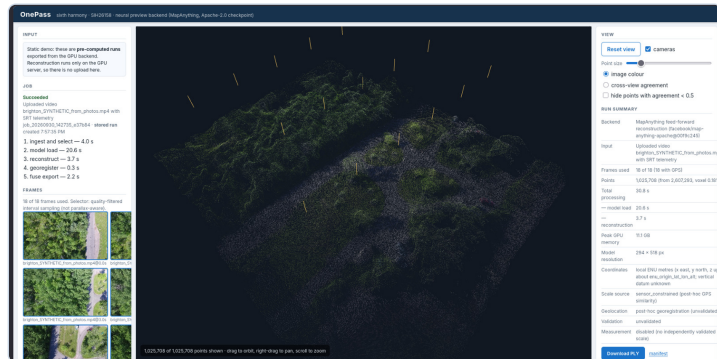
TECHNICAL APPROACH

Methodology and process for implementation (Flow Charts/Images/working prototype)

NEURAL TRACK — IMPLEMENTED, RUNS END TO END **BUILT**

- 1 Ingest** selection.py
ffmpeg decode at a fixed fps; EXIF GPS, or DJI SRT matched by timestamp (≤ 0.5 s gap, no interpolation).
- 2 Select** selection.py
Rejects blur ($< 25\%$ of the set's median Laplacian variance) and bad exposure, then samples evenly to the frame budget.
- 3 Reconstruct** infer.py
MapAnything map-anything-apache @ 00f9c245, bf16 — one feed-forward pass over all views gives poses and depth. No bundle adjustment.
- 4 Georeference** georef.py
Umeyama similarity + RANSAC, camera centres to GPS in local ENU. Refuses near-collinear tracks ($s_2/s_1 < 0.1$).
- 5 Fuse and export** fuse.py
Voxel downsample; binary PLY carrying per-point cross-view agreement, confidence and view id.
- 6 Serve** server.py
FastAPI worker, one cancellable subprocess per job; three.js viewer; hashed run manifests.

MEASURED — A100 80GB, 18 aerial photos: **1,134,767 points in 29.8 s** (18.7 s model load, 5.9 s reconstruct), 11.1 GB peak GPU. Agreement with the photos' own GPS 1.43 m RMSE. Offline run verified in a network-less namespace.



Working prototype: a stored run in the browser viewer — video + SRT, 18 frames, 1,025,708 points, camera poses in gold. Live: bhuvan171.github.io/onepass-neural (ref. 15).

PLANNED — NOT IMPLEMENTED

- Parallax-aware keyframe selection (current selector is interval-based)
- Same frames through the COLMAP track; report inter-method agreement
- Tier 2–3: dense MVS / Gaussian splatting, Poisson mesh
- Moving-object masking; LAS/GLB/OBJ/GeoTIFF export; measurement tools
- Independent accuracy evaluation against surveyed checkpoints

Fully offline and on-premise — no cloud service, no external API. Weights are the **Apache-2.0** MapAnything checkpoint; the CC-BY-NC checkpoint is never downloaded. Two dependencies (plyfile GPL-3.0, system FFmpeg) are still under licence review.

FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

MEASURED ON OUR OWN SYSTEM — 18-photo aerial sample, one A100 (ref. 15)

What we measured	Result	What it does not show
End-to-end reconstruction	1.13 M coloured points in 29.8 s (5.9 s inference)	One sample, and a photo set — not a full-length flight video
Agreement with the photos' own GPS	1.43 m RMSE; 1.62 m leave-one-out	Agreement with input GPS, not ground accuracy. No surveyed checkpoints exist for this sample
Learned metric scale vs GPS	3.96× too small; GPS alignment corrects it	Confirms the model's own scale cannot be trusted on aerial scenes
Degenerate input handling	Single flight line refused ($s_2/s_1 = 0.009$)	Refusal is the correct result, not a reconstruction

NEXT GATE

- A **real** drone video with its SRT through the upload path (only a synthetic video has been tested)
- Same frames through the COLMAP track — inter-method agreement
- Independent accuracy on a benchmark with ground truth (UseGeo, ref. 1–3)

Published work reports newer feed-forward methods leaving ~19% fewer blank patches than the standard method (ref. 1); that study did not use GPS. We have not reproduced it on our system.

RISKS AND MITIGATIONS

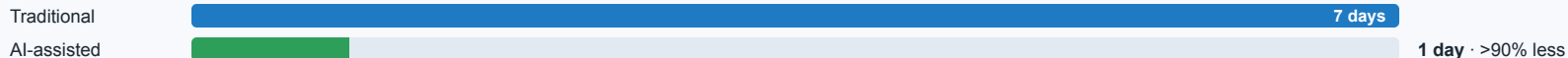
Risk	Mitigation and status
Shape uncertain with one flight path	Refuses to geolocate collinear tracks BUILT
Learned scale unreliable	Post-hoc GPS similarity BUILT
Unseen / occluded surfaces	Per-point cross-view agreement, filterable BUILT
Moving vehicles and people	Segmentation + motion masking PLANNED

IMPACT AND BENEFITS

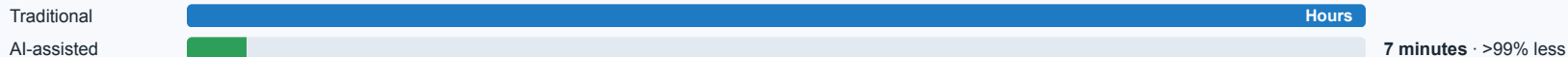
Potential impact on the target audience · Benefits of the solution (social, economic, environmental, etc.)

The bottleneck: time

MOBILIZATION & ASSESSMENT



PROCESSING AT SCALE (2,000 SITES)



Sources: SkyeBrowse (2026) processing benchmark, ref. 14; Texas A&M / Project CLARKE mobilization benchmark, ref. 12 — both external. **Our own measured run:** 18 aerial views → 1.13 M points in 29.8 s on one A100 (ref. 15).

OPERATIONAL

- Removes the ground-marker survey step (ref. 13)
- Runs on-premise and offline, at the point of capture — verified

SECTORS

- Disaster/flood response: 119 drone flights already mapped one hurricane's damage (ref. 12)
- Border mapping, infrastructure, archaeology

ECONOMIC

- Skips a survey crew + RTK-GPS gear, \$1,000–2,000 per site-day (ref. 13)
- No repeat flights, no per-seat licence

ENVIRONMENTAL & SOCIAL

- Fewer flight hours per survey
- Same-day response vs 3–7 days traditionally (ref. 11)

RESEARCH AND REFERENCES

Details / Links of the reference and research work

1. Wu, Landgraf, Ulrich & Qin (2025). *An Evaluation of DUST3R/MAST3R/VGGT 3D Reconstruction on Photogrammetric Aerial Blocks*. arXiv:2507.14798
2. UseGeo benchmark: github.com/3DOM-FBK/UseGeo
3. UseGeo official evaluation code: github.com/UseGeoEvaluation
4. Florea, Miclea & Nedevschi (2021). *WildUAV: Monocular UAV Dataset for Depth Estimation Tasks*. IEEE ICCP: github.com/hrflir/wuav
5. GauU-Scene / U-Scene: large-scale UAV dataset with LiDAR ground truth
6. OpenDroneMap example datasets: github.com/OpenDroneMap/ODMdata
7. *Effective Strategies for Mitigating the Bowl Effect in UAV Photogrammetry for Corridor Projects*. Drones 9(6):387 (2025)
8. Ferrer-González et al. *UAV Photogrammetry Accuracy Assessment for Corridor Mapping*. Remote Sensing 12(15):2447 (2020)
9. Atik (2025). Evaluation of NeRF and Gaussian Splatting point clouds against photogrammetric reference. Sensors 25(10):2995
10. ScienceDirect (2025). *Optimal ground control point layout for UAV photogrammetry in high-precision 3D mapping*.
11. Overwatch Mapping (2026). *Drone response to storms and disasters: how aerial capture compresses the damage-assessment timeline*.
12. DroneLife (2025). *Drone disaster response project CLARKE*, cites CRASAR/Texas A&M's 119-flight Hurricane Harvey deployment.
13. Aerocartwright (2026). *GCP setup for non-surveyors*, ground-control crew and equipment cost estimates.
14. SkyeBrowse (2026). *Drone-to-3D processing time benchmark at scale*.
15. **Team repository and measured runs:** github.com/Bhuvan171/onepass-neural
Live viewer: bhuvan171.github.io/onepass-neural
Every figure on slides 3–5 is reproduced in *BUILD_STATUS.md*; permitted claims and their limits in *CLAIMS.md*.